

Justify your answers and show your work.

Complete the following steps for $f(x) = 24 - 2x$, interval $[0, 6]$, and $n = 3$.

1. Sketch the graph of the function on the given interval and illustrate the *right Riemann sums* using n subintervals.

2. Calculate the *right Riemann sums* for the given interval and number of subintervals, n .

The figure shows the area of regions bounded by the graph of $f(x)$. Evaluate the following integrals.

3. $\int_0^8 f(x) dx$

4. $\int_0^5 f(x) dx - \int_8^0 f(x) dx$

Evaluate the following indefinite integrals and definite integrals.

5. $f(x) = 9x^8$

a. $\int 9x^8 dx$

b. $\int_0^1 9x^8 dx$

6. $f(x) = x^2 - 8x + 1$

a. $\int (x^2 - 8x + 1) dx$

7. $f(x) = \cos 3x$

a. $\int \cos 3x dx$

b. $\int_{-\pi/3}^{\pi/3} \cos 3x dx$

8. $f(x) = \frac{x - \sqrt{x}}{x^3}$

a. $\int \frac{x - \sqrt{x}}{x^3} dx$

9. $f(x) = x^2(3x^3 + 1)^4$

a. $\int x^2(3x^3 + 1)^4 dx$

b. $\int_1^2 x^2(3x^3 + 1)^4 dx$

10. $v(\theta) = \cos \theta \sin^2 \theta$

a. $\int \cos \theta \sin^2 \theta d\theta$

11. $g(t) = \frac{t}{\sqrt{4-t^2}}$
a. $\int \frac{t}{\sqrt{4-t^2}} dt$

b. $\int_2^4 \frac{t}{\sqrt{4-t^2}} dt$

12. A hiking trail has an elevation given by $h(x) = 40x^3 - 300x^2 + 200x + 4000$ where h measures feet above sea level and x represents miles along the trail with $0 \leq x \leq 5$. What is the average elevation along the trail?

13. A bowling ball rolls down a ramp at a speed of $v(t) = -3t^2$ feet per second (where t is seconds). The distance of the ball from the top of the ramp is 4 feet after 2 seconds. Find the function giving the distance of the ball from the top of the ramp at t seconds.